DIPLOMA PROGRAMME PRACTICE · HL · SET B

Chemistry / Paper 1A

Question booklet · 40 marks · 120 minutes across 1A and 1B

Answer every question. Choose one option, A–D, per question. Use a calculator and a clean Chemistry data booklet. Record choices online or on separate paper.

QUESTION 1 [1]

An ion has mass number 52, 28 neutrons and 21 electrons. Which change gives a neutral atom of the same isotope?

A Gain three electrons
B Lose three electrons
C Lose seven neutrons
D Gain three protons

QUESTION 2 [1]

A mass spectrum contains singly charged atomic ions of masses 24, 25 and 26. Mass 24 accounts for 60.0% and the relative atomic mass is 24.65. What is the percentage abundance of mass 26?

A 32.5%
B 15.0%
C 35.0%
D 25.0%

QUESTION 3 [1]

Which species has the greatest number of unpaired electrons in its gaseous ground state? Use atomic numbers N 7, O 8 and F 9.

A F
B O
C N
D F^-

QUESTION 4 [1]

A convergence photon has energy 8.20×10^{-19} J per atom. What is the corresponding molar ionization energy, using $N_A=6.02\times10^{23}$ mol^{-1}?

A 4.94×10^{5} kJ mol^{-1}
B 0.494 kJ mol^{-1}
C 494 kJ mol^{-1}
D 49.4 kJ mol^{-1}

QUESTION 5 [1]

Which explanation accounts for both first-ionization-energy decreases Mg→Al and P→S across period 3?

A Both decreases result from a decrease in nuclear charge

B Al and S both begin a new principal shell

C Al loses a 3p electron; S loses one of a paired set of 3p electrons

D Al loses a 3s electron; S has no paired 3p electrons

QUESTION 6 [1]

At the same temperature and pressure, 0.300 g of gas A occupies 150 cm^3 and 0.400 g of gas B occupies 100 cm^3. Both are ideal. What is M(B)/M(A)?

A 0.750

B 2.00

C 1.33

D 0.500

QUESTION 7 [1]

A solid containing only Mg^{2+} and N^{3-} shatters when layers are displaced. Which pair gives its formula and the cause of fracture?

A Mg_3N_2; like charges are brought together

B MgN; a sea of electrons becomes immobile

C Mg_2N_3; unlike charges repel

D Mg_3N_2; neutral molecules separate

QUESTION 8 [1]

Which molecule has polar bonds whose dipoles cancel because of its molecular geometry?

A BF_3

B NF_3

C SO_2

D CH_3F

QUESTION 9 [1]

The central atoms in $PCl_5(g)$ and ICl_4^- have how many lone pairs, respectively?

A 1 and 2

B 0 and 1

C 1 and 1

D 0 and 2

QUESTION 10 [1]

For linear N–N–O, consider the contributor N≡N–O with terminal N having one lone pair and O three. Which assignment gives the formal charges in that order?

A 0, +1, −1
B 0, 0, 0
C +1, 0, −1
D −1, +1, 0

QUESTION 11 [1]

In CH_2=CH–C(=O)–CH_3, which statement is correct about the four carbon atoms and the pi bonds?

A Three carbons are sp^2 and there are two pi bonds
B Two carbons are sp^2 and there are three pi bonds
C One carbon is sp and there is one pi bond
D All four carbons are sp^2

QUESTION 12 [1]

The repeat of poly(propene) is C_3H_6. Ignore end groups. A chain has relative mass 42,000, and C=12, H=1. How many backbone carbon atoms does it contain?

A 3000
B 2000
C 6000
D 1000

QUESTION 13 [1]

A polyamide has a repeat –NH–$(CH_2)_5$–C(=O)–. Which single monomer could produce it by condensation?

A $H_2N(CH_2)_5CH_3$
B $HO(CH_2)_5COOH$
C $H_2N(CH_2)_6NH_2$
D $H_2N(CH_2)_5COOH$

QUESTION 14 [1]

In the reaction $Cl_2+2Br^-\rightarrow 2Cl^-+Br_2$, what happens to the reducing agent?

A Its oxidation state decreases from 0 to −1
B It remains as chloride ions
C It accepts two electrons per atom
D Its oxidation state increases from −1 to 0

QUESTION 15 [1]

Two aqueous ions have configurations [Ar]3d^{10} and [Ar]3d^{9}. Under the simple d–d model, which prediction follows?

A d^{9} cannot absorb because every d orbital contains an electron
B Both must absorb identical wavelengths
C Visible d–d absorption is possible for d^{9}; the filled d^{10} subshell lacks the needed vacancy
D Only d^{10} absorbs because it has more electrons

QUESTION 16 [1]

A sample labelled 'butan-2-ol' is converted to a ketone without carbon–carbon bond cleavage. Which statement about the starting material and product is correct?

A The alcohol loses two H atoms overall and its four-carbon skeleton is retained
B Both have molecular formula $C_4H_{10}O$
C The product is a carboxylic acid
D The alcohol gains H_2 and gives butanal

QUESTION 17 [1]

A saturated ester with formula $C_4H_8O_2$ shows two 3H signals and one 2H signal. The 2H quartet is near 4.1 ppm and one 3H singlet is near 2.0 ppm. Which compound fits?

A Propan-2-yl methanoate
B Ethyl methanoate
C Ethyl ethanoate
D Methyl propanoate

QUESTION 18 [1]

A molecular ion gives singly charged peaks at m/z 108 and 110 of nearly equal intensity. Both can produce an ion at m/z 29. Which assignment fits the pattern and both neutral losses?

A C_2H_5Br yields $C_2H_5^+$ by loss of HBr
B C_2H_5Br yields $C_2H_5^+$ by loss of a Br atom
C C_2H_5Cl yields $C_2H_5^+$ by loss of a Cl atom
D C_2H_5Br yields Br^+ at m/z 29

QUESTION 19 [1]

Which two compounds can each exhibit geometrical isomerism about a C=C bond?

A Ethene and propene
B But-2-ene and 1,2-dichloroethene
C But-1-ene and but-2-ene
D 2-methylpropene and 1,2-dichloroethene

QUESTION 20 [1]

A burner transfers heat to 150 g of water. Its temperature rises by 8.0 K. Use c=4.18 J g^{-1} K^{-1}. The burner loses 0.400 g of fuel. What is the measured energy transferred per gram burned?

A 2.01 kJ g^{-1}

B 50.2 kJ g^{-1}

C 5.02 kJ g^{-1}

D 12.5 kJ g^{-1}

QUESTION 21 [1]

For solid MBr, $\Delta H_f^\circ = -350$, atomization of M=+110, conversion of ½Br_2(l) to Br(g)=+112, ionization of M=+495 and addition of an electron to Br(g)=−325, all in kJ mol^{-1}. What is the lattice formation enthalpy?

A −742 kJ mol^{-1}

B +742 kJ mol^{-1}

C −92 kJ mol^{-1}

D −417 kJ mol^{-1}

QUESTION 22 [1]

Combustion enthalpies of C(graphite), H_2(g) and CH_3OH(l) are −394, −286 and −726 kJ mol^{-1}, all to CO_2(g) and H_2O(l) as applicable. What is ΔH_f° of CH_3OH(l)?

A −240 kJ mol^{-1}

B +240 kJ mol^{-1}

C −46 kJ mol^{-1}

D −1692 kJ mol^{-1}

QUESTION 23 [1]

Fuel P releases 50 MJ kg^{-1} and makes 2.5 kg CO_2 per kg. Fuel Q releases 30 MJ kg^{-1} and makes 1.8 kg CO_2 per kg. At equal efficiency, which statement is correct?

A Q releases less CO_2 per MJ

B A comparison is impossible without molar masses

C P releases less CO_2 per MJ

D Both give 0.050 kg CO_2 per MJ

QUESTION 24 [1]

For 2A(g)→B(g), S°(A)=180 and S°(B)=245 J K^{-1} mol^{-1}. If $\Delta H^\circ = -32.0$ kJ per reaction as written, which is correct at 300 K?

A $\Delta G^\circ = +2.50$ kJ; forward is not spontaneous under standard conditions

B $\Delta G^\circ = -2.50$ kJ; forward is spontaneous

C $\Delta G^\circ = -66.5$ kJ; forward is spontaneous

D $\Delta G^\circ = +34.5$ kJ; forward is not spontaneous

QUESTION 25 [1]

A reaction has K=25.0 at 310 K. What is ΔG°, using R=8.31 J mol⁻¹ K⁻¹?

A +8.29 kJ mol⁻¹
B −8.29 kJ mol⁻¹
C −64.4 kJ mol⁻¹
D −3.60 kJ mol⁻¹

QUESTION 26 [1]

A tablet contains 0.840 g $NaHCO_3$ (M=84.0) and an inert binder. What volume of 0.250 mol dm⁻³ HCl reacts exactly with the bicarbonate?

A 80.0 cm³
B 40.0 cm³
C 20.0 cm³
D 10.0 cm³

QUESTION 27 [1]

At fixed temperature, halving [P] while doubling [Q] leaves the rate unchanged. Doubling [Q] alone doubles the rate. What is the predicted factor when [P] triples and [Q] halves?

A 6.0
B 1.5
C 0.75
D 3.0

QUESTION 28 [1]

For a two-step mechanism the energies of reactants, first peak, intermediate, second peak and products are 0, 55, −15, 50 and −30 kJ mol⁻¹. Which elementary forward activation energy is larger?

A Step 1 by 10 kJ mol⁻¹
B Step 1 by 5 kJ mol⁻¹
C Step 2 by 10 kJ mol⁻¹
D Step 2 by 65 kJ mol⁻¹

QUESTION 29 [1]

A rate constant doubles between 300 and 315 K. With R=8.31 J mol⁻¹ K⁻¹, which Ea follows from the Arrhenius relationship?

A 0.384 kJ mol⁻¹
B 36.3 kJ mol⁻¹
C −36.3 kJ mol⁻¹
D 18.2 kJ mol⁻¹

QUESTION 30 [1]

A sealed equilibrium A(g) ⇌ 2B(g) has a forward exothermic reaction. Which single change increases the equilibrium amount of B?

A Add a catalyst at fixed temperature
B Remove A at fixed volume and temperature
C Increase the temperature at fixed volume
D Expand the container at fixed temperature

QUESTION 31 [1]

For $2HI \rightleftharpoons H_2+I_2$, 1.00 mol HI is placed in 2.00 dm^3. At equilibrium 0.200 mol H_2 has formed. What is Kc?

A 9.00
B 0.0625
C 0.250
D 0.111

QUESTION 32 [1]

For A(g)+2B(g) ⇌ C(g), the volume of an equilibrium mixture is halved at constant T. What is Qc immediately after compression?

A 2Kc
B Kc/4
C 4Kc
D Kc/2

QUESTION 33 [1]

Mix 20.0 cm^3 of 0.100 mol dm^{-3} HCl with 10.0 cm^3 of 0.100 mol dm^{-3} NaOH. Assuming additive volumes, what is the pH?

A 2.00
B 7.00
C 1.00
D 1.48

QUESTION 34 [1]

A 20.0 cm^3 sample of weak monoprotic acid is titrated to equivalence with 32.0 cm^3 NaOH. The pH at 16.0 cm^3 added is 4.65. What is Ka?

A 4.47×10^{4}
B 4.65×10^{-5}
C 2.24×10^{-5}
D 2.24×10^{-10}

QUESTION 35 [1]

At 298 K a base solution has pH 11.30. Which concentration pair is consistent, in mol dm^{-3}? Use $K_w=1.00\times10^{-14}$.

A $[OH^-]=5.01\times10^{-12}$ and $[H^+]=2.00\times10^{-3}$

B $[OH^-]=2.00\times10^{-2}$ and $[H^+]=5.01\times10^{-13}$

C $[OH^-]=2.00\times10^{-3}$ and $[H^+]=5.01\times10^{-12}$

D $[OH^-]=2.70$ and $[H^+]=11.30$

QUESTION 36 [1]

For reduction couples X^{2+}/X and Y^{2+}/Y, E° values are +0.25 V and −0.50 V. Which result follows for $2Y(s)+2X^{2+}\rightarrow2Y^{2+}+2X(s)$ under standard conditions?

A E°=−0.75 V, with four electrons transferred

B E°=+0.75 V, with two electrons transferred per equation as written

C E°=+1.50 V, with four electrons transferred

D E°=+0.75 V, with four electrons transferred per equation as written

QUESTION 37 [1]

When dilute aqueous sodium sulfate is electrolysed with inert electrodes, which products are expected in the usual water-electrolysis model?

A H_2 at the cathode and O_2 at the anode

B Na at the cathode and S at the anode

C O_2 at the cathode and H_2 at the anode

D Na at the cathode and SO_2 at the anode

QUESTION 38 [1]

During substitution of methane with bromine, which pair of elementary steps gives the net propagation reaction $CH_4+Br_2\rightarrow CH_3Br+HBr$?

A $Br\bullet+CH_4\rightarrow HBr+CH_3\bullet$ and $CH_3\bullet+Br\bullet\rightarrow CH_3Br$

B $Br_2\rightarrow2Br\bullet$ and $2Br\bullet\rightarrow Br_2$

C $Br\bullet+CH_4\rightarrow HBr+CH_3\bullet$ and $CH_3\bullet+Br_2\rightarrow CH_3Br+Br\bullet$

D $CH_3\bullet+CH_3\bullet\rightarrow C_2H_6$ and $Br_2\rightarrow2Br\bullet$

QUESTION 39 [1]

Which change best supports an SN1 pathway for a tertiary bromoalkane undergoing hydrolysis?

A Rate doubles on doubling either concentration

B Rate is independent of substrate concentration

C Only a single concerted transition state is observed

D Rate doubles on doubling substrate concentration and is unchanged on doubling nucleophile concentration

QUESTION 40 [1]

In electrophilic addition of HBr to an alkene, the first curly arrow for protonation should start and end where?

A From the H atom to the pi bond

B From the alkene pi bond to the H atom of HBr

C From a carbon nucleus to bromine

D From Br^- to the alkene before polarization